



Control Statements (Loops) Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Arrays.

Section 1: Loop Fundamentals (for, while, do-while)

★ Print Numbers from 1 to N using for Loop

★ Print Numbers from N to 1 using for Loop

★ Print Numbers using while Loop

★ Print Numbers using do-while Loop

Print Even Numbers up to N

Print Odd Numbers up to N

Count Numbers from A to B

Multiples of N

Reverse Counting

Skip Multiples of 3

Section 2: Summation Problems

★ Sum of First N Numbers

★ Sum of Even Numbers

★ Sum of Odd Numbers

Average of N Numbers

Sum of Multiples of 3 and 5

Product of First N Numbers

Running Total Calculator

Section 3: Digit-Based Problems

★ Count Digits

★ Sum of Digits

★ Product of Digits

★ Reverse a Number

First Digit of a Number

Last Digit of a Number

Largest Digit

Smallest Digit



Problem Statements

Even Digit Count

Odd Digit Count

Control Statement (Java)

Section 4: Number Properties

- ★ **Palindrome Number**
- ★ **Armstrong Number**
- ★ **Prime Number Check**
- ★ **Perfect Number**
- ★ **Strong Number**
 - Harshad Number
 - Neon Number
 - Duck Number
 - Automorphic Number

Section 5: Number Series

- ★ **Fibonacci Series**
 - Arithmetic Progression
 - Geometric Progression
 - Square Series
 - Cube Series
 - Power Series

Section 6: Factor Problems

- ★ **Factorial**
- ★ **Factors of a Number**
- ★ **Count Factors**
- ★ **Sum of Factors**
- ★ **Prime Factors**
 - Common Factors of Two Numbers

Section 7: Range Problems

- ★ **Prime Numbers in Range**
- ★ **Armstrong Numbers in Range**
 - Palindrome Numbers in Range
 - Perfect Numbers in Range
 - Strong Numbers in Range
 - Even Numbers in Range



Section 8: GCD and LCM

★ GCD of Two Numbers

★ LCM of Two Numbers

Co-Prime Check

HCF using Factors

Section 9: Pattern Fundamentals

★ Square Pattern

★ Right Triangle Pattern

★ Inverted Triangle Pattern

★ Number Triangle Pattern

★ Floyd's Triangle

Alphabet Triangle

Section 10: Intermediate Patterns

★ Pyramid Pattern

★ Inverted Pyramid Pattern

★ Diamond Pattern

Hollow Square

Hollow Rectangle

X Pattern

Butterfly Pattern

Section 11: Labeled Loops and Loop Control (Java-Specific)

★ Break Statement in Nested Loop

★ Continue Statement in Nested Loop

★ Labeled Break in Nested Loop

★ Labeled Continue in Nested Loop

Find First Matching Pair using Labeled Break

Skip Outer Loop Iteration using Labeled Continue

Controlled Infinite Loop with Break Condition

Section 12: Enhanced For-Loop / for-each Problems (Java-Specific)

★ Iterate Array using Enhanced For-Loop

★ Sum of Array Elements using Enhanced For-Loop

★ Find Max and Min in Array using Enhanced For-Loop

★ Print 2D Array using Nested Enhanced For-Loop



Problem Statements

Count Elements Matching a Condition using for-each

Control Statement (Java)

Section 13: Real World Loop Applications

- ★ **Multiplication Table Generator**
- ★ **ATM Transaction History Simulation**
- ★ **Student Marks Analyzer**
- ★ **Electricity Bill Summary for N Customers**
- ★ **Employee Salary Report**

Menu Driven Calculator using Loop